Rajkumar Rathor
Software Engineer Intern

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Summary

Final-year B.Tech Computer Science student (Expected 2027) with hands-on experience in Python development, machine learning, and frontend web technologies. Built and deployed ML models for real-world classification and prediction tasks. Proficient in writing clean, efficient code across the full development stack. Actively seeking a Software Engineer internship to apply and grow technical skills in a fast-paced environment.

Education

Bachelor of Technology — Computer Science Engineering | **Expected Graduation: 2027**
Jaypee University of Engineering and Technology, Guna | CGPA: 6.33 / 10

CBSE Class 12th — Senior Secondary | **Completed**
Senior Secondary Education | 67%

Technical Skills

Programming Languages: C, C++, Python, JavaScript
Web Development: HTML, CSS, JavaScript (Responsive Design, UI/UX Basics)
Machine Learning: Supervised Learning, Classification, Regression, Scikit-learn, Random Forest, SVM, Naive Bayes, LightGBM
Data & NLP: Pandas, NumPy, TF-IDF, Feature Engineering, Data Cleaning, Natural Language Processing (NLP)
Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code
Core CS: Data Structures & Algorithms (DSA), OOP, Graph Algorithms (BFS, DFS), 350+ LeetCode Problems Solved

Experience

Software Developer Intern — Python | **Sep 2025 – Oct 2025**
CodeAlpha (Remote) | Python — OOP — Automation — Debugging

- Built 3+ Python applications applying OOP principles, modular design, and file handling within a 1-month remote internship.
- Automated repetitive data processing tasks using Python scripts, cutting manual effort by ~40%.
- Improved software reliability through systematic debugging, code reviews, and adherence to clean-code best practices.
- Strengthened problem-solving skills by implementing algorithms and data structures (lists, dictionaries, functions) in real projects.

Projects

Email Spam Detection System | *Python, NLP, Machine Learning* | **Jan 2026 – Apr 2026**

- Engineered an NLP-based spam classifier achieving **96%+ accuracy** on a labeled dataset of **5,500+ emails**.
- Built a full text preprocessing pipeline — stopword removal, stemming, TF-IDF vectorization — for robust feature extraction.
- Trained and benchmarked 4 ML models (Logistic Regression, SVM, Naive Bayes, LightGBM); selected best via F1-score analysis.
- Reduced false positive rate by ~20% through systematic evaluation using precision, recall, and confusion matrix metrics.

Traffic Flow Prediction System | *Python, Machine Learning* | **Aug 2025 – Oct 2025**

- Developed a supervised ML model to predict traffic patterns using historical data and road condition features.
- Designed an end-to-end preprocessing pipeline (data cleaning, feature selection, normalization) boosting accuracy by ~15%.
- Compared Linear Regression and Random Forest models; Random Forest reduced RMSE by ~12% over baseline.
- Extracted actionable insights from peak-hour and traffic density analysis to support real-time decision-making.

Certifications

- **Infosys Springboard — Python Programming:** Core Python including OOP, file handling, loops, functions, and problem-solving.
- **AlgoUniversity — Graph Algorithms:** Graph theory covering BFS, DFS, shortest path, and traversal techniques.